

Technical Synthesis: Audio & Data Kill-Gate

This document integrates findings from the ADC24 sessions and native hardware crates to define the production-ready stack for the Acoustic Claw.

1. Native Audio Capture (Hardware Access)

To pass the Kill-Gate on 2GB RAM devices, we move away from high-level abstractions to platform-specific drivers.

Android Target (Dakar Pilot)

- **Primary: oboe-rs**
 - **Logic:** Wraps Google's Oboe C++. It is the standard for low-latency Android audio.
 - **Capability:** Automatically switches between AAudio (modern) and OpenSL ES (legacy).
 - **Kill-Gate Feature:** Use `SharingMode::Exclusive` for the lowest possible latency by bypassing the system mixer.
- **Support: android-media**
 - **Logic:** Provides NDK bindings for low-level media tasks.
 - **Use Case:** Direct hardware-accelerated tokenization if CPU cycles become a bottleneck for FocalCodec.

iOS/macOS Target (NYC Pilot)

- **Primary: coreaudio-rs**
 - **Logic:** Provides low-level access to Apple's CoreAudio framework.
 - **Use Case:** High-fidelity capture for iPhone users during the World Cup.

Unified Fallback Logic

1. **Try oboe-rs** (AAudio mode).
2. **Fallback to oboe-rs** (OpenSL ES mode) if AAudio is unavailable.
3. **Final Fallback to cpal** for generic device support.

2. Thread Synchronization (Audio-to-Inference)

Based on the **ADC-Audiodev-slides.pdf**, we avoid Mutexes in the audio callback to prevent priority inversion.

- **Audio Data Transfer:** Use `rtrb` (real-time ringbuffer) or `ringbuf`. These are lock-free, single-producer/single-consumer buffers optimized for audio streams.
- **Message Passing:** Use `crossbeam_channel::bounded` for non-blocking communication between the audio thread and the **YAATAL Reactor**.

- **Parameter Management:** Use `atomic_float::AtomicF32` for real-time gain/threshold adjustments from the UI without locking the audio thread.

3. Data Persistence & Sync (The Ledger)

Based on the **ADC-Synchro-data** PDF, the project requires an offline-first strategy for unstable network conditions in Dakar.

- **The Choice: libSQL (Turso)**
 - **Why:** It is a Rust-native fork of SQLite.
 - **Feature:** Includes a two-way synchronization feature (in beta) specifically designed for edge-to-cloud sync.
 - **Integration:** Replaces proprietary solutions (Realm/Couchbase) with a fully open-source, local-first database that runs natively on the phone.
- **Reasoning Injection:** Use **libSQL** to store the 26-byte TinyLoRA payloads. When a new logic update is pushed, the database syncs, and the engine hot-reloads the parameters from the local file.

4. Kill-Gate Execution Plan

Component	Implementation	Success Metric
Audio Capture	oboe-rs (Exclusive)	Sub-10ms buffer latency.
Data Buffer	rtrb	Zero dropped samples between hardware and LTC kernel.
Persistence	libSQL	Local write success with 0% network connectivity.
Sync	Turso Beta	Automatic background reconciliation when 2G/LTE signal returns.